

Elastic Beanstalk

Class no:25

Elastic Beanstalk

Think of AWS Elastic Beanstalk as the "easy button" for deploying web applications. If you're a developer who wants to get your code onto the internet without spending hours configuring servers, load balancers, and scaling rules, this is your go-to service.

AWS Elastic Beanstalk: Overview

1. Definition

AWS Elastic Beanstalk is a **Platform as a Service (PaaS)** that allows for the quick deployment and management of applications in the AWS Cloud. You simply upload your code, and Elastic Beanstalk automatically handles the deployment details of capacity provisioning, load balancing, auto-scaling, and application health monitoring.

2. Key Components

- **Application:** A logical collection of Elastic Beanstalk components (environments, versions, and configurations).
- **Application Version:** A specific, labeled iteration of your deployable code (usually a Java .war file, a zip file, or a Docker image).
- **Environment:** A version that is deployed onto AWS resources.
 - *Web Server Environment:* Handles HTTP requests (standard web apps).
 - *Worker Environment:* Handles background tasks or long-running processes via an SQS queue.
- **Environment Tier:** Determines whether the environment runs a web app or a background task.

3. Supported Platforms

Elastic Beanstalk supports a wide variety of languages and platforms:

- **Programming Languages:** Java, .NET, PHP, Node.js, Python, Ruby, and Go.
- **Containers:** Docker.
- **Web Servers:** Passenger, Apache, Nginx, and IIS.

4. Why Use It? (Benefits)

- **Speed:** Go from code to a live URL in minutes.
- **Developer Productivity:** You manage the code; AWS manages the infrastructure.
- **Full Control:** While it automates the setup, you still have "root" access to the underlying AWS resources (like EC2 instances) if you need to tweak them.
- **Cost:** There is **no additional charge** for Elastic Beanstalk itself. You only pay for the underlying AWS resources (EC2, S3, etc.) that your application uses.

5. How It Works (The Workflow)

1. **Create Application:** Define your app name.
2. **Upload Version:** Bundle your code and upload it.
3. **Launch Environment:** AWS provisions the resources (EC2, Load Balancers, etc.).
4. **Manage:** Use the dashboard to monitor health, view logs, or update the application.

6. Deployment Strategies

- **All at Once:** Fast, but involves downtime.
- **Rolling:** Replaces instances in "batches" to avoid total downtime.
- **Immutable:** Creates a whole new set of instances to ensure the new version works before killing the old ones.
- **Blue/Green:** Swaps the CNAME of two separate environments for zero-downtime transitions.

Note: Elastic Beanstalk is perfect for startups and enterprise teams who want to stay focused on writing features rather than managing the "plumbing" of the cloud.



Fig.a. Components of Elastic Beanstalk

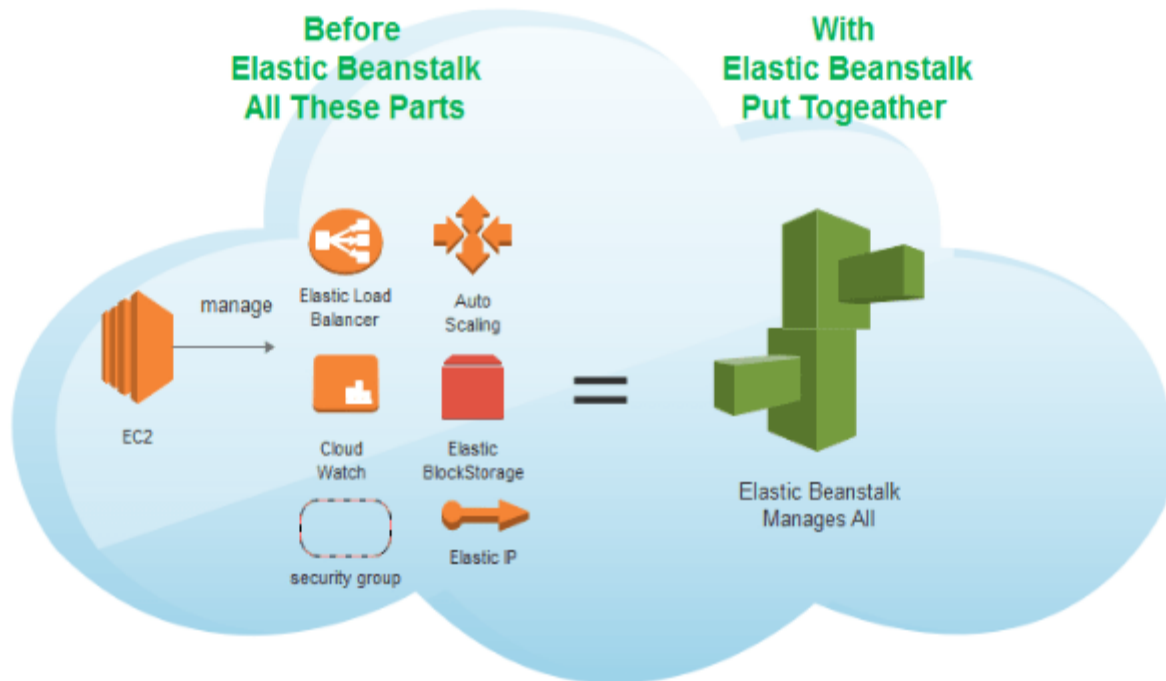
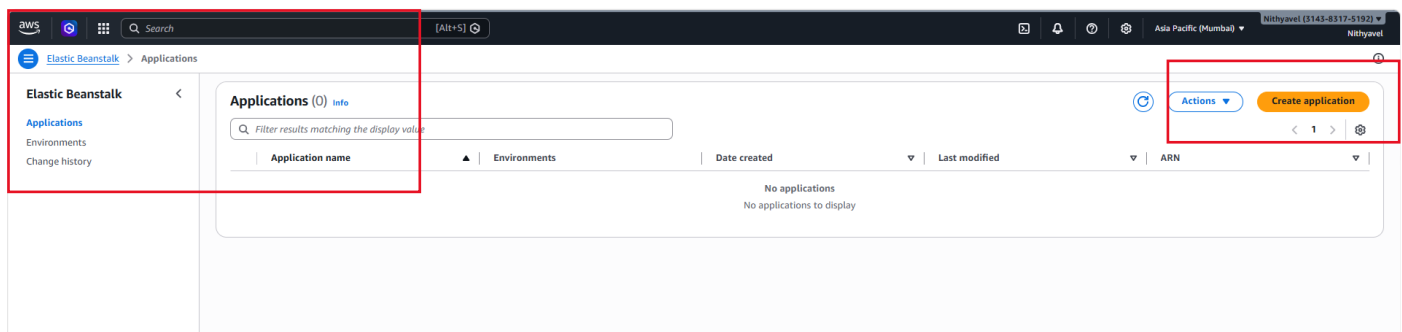


Fig.b.How Elastic BeanStalk works.

Hands on: How to Elastic Beanstalk and a EC2 instance for it.

Step no:1 Search Elastic Beanstalk → Application → Create Application.



Step no:2 Provide a name: Myapplication → Create.

Create new application Info

Application information

Application name

Myapplication

Maximum length of 100 characters.

Description

for practice

Tags

Apply up to 50 tags. You can use tags to group and filter your resources. A tag is a key-value pair. The key must be unique within the resource and is case-sensitive. [Learn more](#)

No tags associated with the resource.

Add new tag

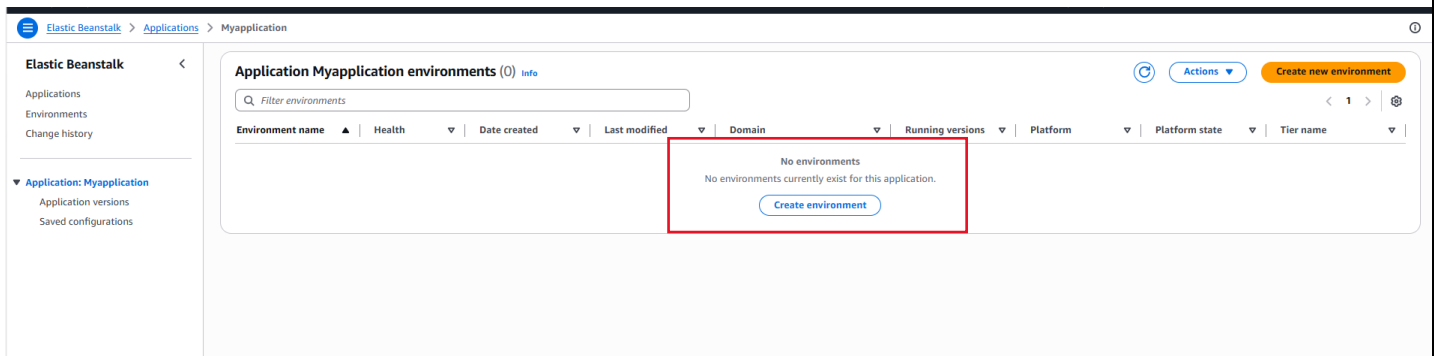
You can add up to 50 tags.

Cancel

Create

Step no:3 It will create an application. Now **Create new environment**.

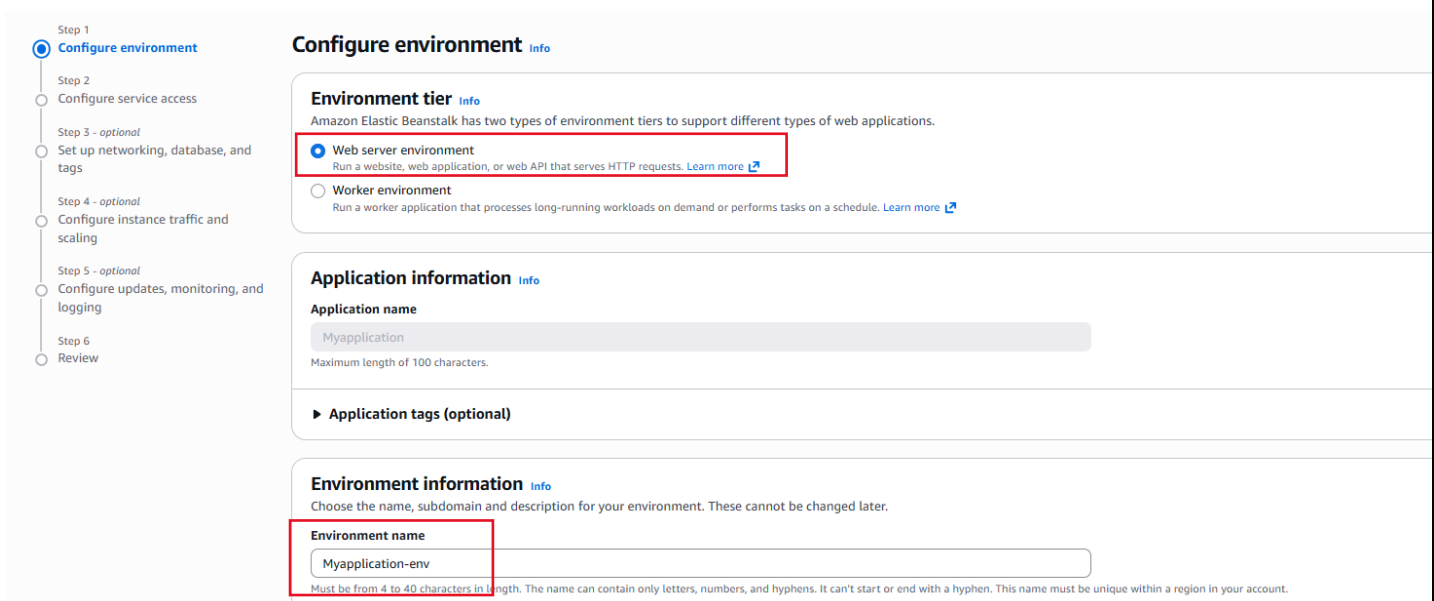
(We can create multiple environments for our application like development, staging and production environment).



Step no:4 Now configure the environment,

Tier: **Web server environment**

Environment name: **Myapplication-env**



Leave the domain name as blank, it will create a new one for us.

Platform: **Java**

Application code: **Sample application**

(We are just testing so use the existing code, there is also option to upload our code if we are deploying our own application)

Presets: Choose the free tier eligible.

Next.

Domain
.ap-south-1.elasticbeanstalk.com

▶ Environment description

Platform Info

Platform
Java
Platform branch
Corretto 25 running on 64bit Amazon Linux 2023
Platform version
4.11.1 (Recommended)

Application code Info

☒ Sample application
☐ Existing version
Application versions that you have uploaded.
☐ Upload your code
Upload a source bundle from your computer or copy one from Amazon S3.

Presets Info

Start from a preset that matches your use case or choose custom configuration to unset recommended values and use the service's default values.

Configuration presets
☒ Single instance (free tier eligible)
☐ Single instance (using spot instance)
☐ High availability
☐ High availability (using spot and on-demand instances)
☐ Custom configuration

Step no:5 For service role → click **Create role**. It will have a respective configured role.

As we know in order to connect a service (EC2) with another service(EBS) we need a role (IAM).

Step 1
Configure environment
Step 2
Configure service access
Step 3 - optional
Set up networking, database, and tags
Step 4 - optional
Configure instance traffic and scaling
Step 5 - optional
Configure updates, monitoring, and logging
Step 6
Review

Configure service access Info

Service access
IAM roles, assumed by Elastic Beanstalk as a service role, and EC2 instance profiles allow Elastic Beanstalk to create and manage your environment. Both the IAM role and instance profile must be attached to IAM managed policies that contain the required permissions. [Learn more](#)

Service role
Choose an IAM role for Elastic Beanstalk to assume as a service role. The IAM role must have the required IAM managed policies.

EC2 instance profile
Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.

EC2 key pair - optional
Select an EC2 key pair to securely log in to your EC2 instances. [Learn more](#)

We can see the service use case as Elastic beanstalk (env) → next.

Select trusted entity Info

Trusted entity type
☒ AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
☐ AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
☐ Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
☐ SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
☐ Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.
Service or use case
Elastic Beanstalk
Choose a use case for the specified service.
Use case
☐ Elastic Beanstalk - Compute
Allows your environment's EC2 instances to perform operations required for your application.
☒ Elastic Beanstalk - Environment
Allows access to other AWS service resources that are required to create and manage environments.

We can see it has automatically add the required permissions and policies → next.

Add permissions [Info](#)

Permissions policies (2) [Info](#)

The type of role that you selected requires the following policy.

Policy name [i](#)

▲ | Type

▼ |

 AWSElasticBeanstalkEnhancedHealth	AWS managed
 AWSElasticBeanstalkManagedUpdatesCustomerRolePolicy	AWS managed

► Set permissions boundary - *optional*

Cancel

Previous

Next

Review → Create role.

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and "+, -, @, _" characters.

Description

Add a short explanation for this role.

Maximum 1000 characters. Use letters [A-Z and a-z], numbers [0-9], tabs, new lines, or any of the following characters: "+, -, @, _" [0-9A-Za-z_+@-].

Step 1: Select trusted entities

Edit

Trust policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "Service": "elasticbeanstalk.amazonaws.com"
8       },
9       "Action": "sts:AssumeRole"
10    }
11  ]
12 }
```

Step 2: Add permissions

Edit

Permissions policy summary

Policy name [i](#)

▲ | Type

▼ | Attached as

▼

AWSElasticBeanstalkEnhancedHealth	AWS managed	Permissions policy
AWSElasticBeanstalkManagedUpdatesCustomerRolePolicy	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional [Info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

Cancel

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Create role

Once created → do to our EBS page → refresh → we can see our own created role: **aws-elasticbeanstalk-service-role**.

Service role

Choose an IAM role for Elastic Beanstalk to assume as a service role. The IAM role must have the required IAM managed policies.

Choose a instance profile

Similarly, **Ec2 instance profile**.

Choose a new / existing keypair.

Configure service access [Info](#)

Service access

IAM roles, assumed by Elastic Beanstalk as a service role, and EC2 instance profiles allow Elastic Beanstalk to create to IAM managed policies that contain the required permissions. [Learn more](#)

Service role

Choose an IAM role for Elastic Beanstalk to assume as a service role. The IAM role must have the required IAM managed policies.

aws-elasticbeanstalk-service-role

EC2 instance profile

Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.

aws-elasticbeanstalk-ec2-role

EC2 key pair - optional

Select an EC2 key pair to securely log in to your EC2 instances. [Learn more](#)

mumkp

Step no:6 Choose default VPC,

Enable Public IP

AZs: Choose all for more availability

Rest as default

Next.

- Step 2
Configure service access
- Step 3 - optional
Set up networking, database, and tags
- Step 4 - optional
Configure instance traffic and scaling
- Step 5 - optional
Configure updates, monitoring, and logging
- Step 6
Review

Instance settings

Choose a subnet in each AZ for the instances that run your application. To avoid exposing your instances to the Internet, run your instances in private subnets and load balancer in public subnets. To run your load balancer and instances in the same public subnets, assign public IP addresses to the instances. [Learn more](#)

VPC

Launch your environment in a custom VPC instead of the default VPC. You can create a VPC and subnets in the VPC management console. [Learn more](#)

vpc-0538c231a226fe3c9 | (172.31.0.0/16)



Create VPC

Public IP address

Assign a public IP address to the Amazon EC2 instances in your environment.

☒ Enable

Instance subnets

Filter instance subnets

☒	Availability Zone	Subnet	CIDR	Name
☒	ap-south-1a	subnet-01bf4d440c0912db3	172.31.32.0/20	
☒	ap-south-1b	subnet-0b3c21b6de4314d95	172.31.0.0/20	
☒	ap-south-1c	subnet-0c305b80f72541592	172.31.16.0/20	

Database [Info](#)

Integrate an RDS SQL database with your environment. [Learn more](#)

☐ Enable database

Tags

Apply up to 50 tags. You can use tags to group and filter your resources. A tag is a key-value pair. The key must be unique within the resource and is case-sensitive. [Learn more](#)

Cancel

Skip to Review

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Next

Choose the Security group as **default SG**.

- Step 1
Configure environment
- Step 2
Configure service access
- Step 3 - optional
Set up networking, database, and tags
- Step 4 - optional
Configure instance traffic and scaling
- Step 5 - optional
Configure updates, monitoring, and logging
- Step 6
Review

Configure instance traffic and scaling - optional [Info](#)

Instances [Info](#)

Configure the Amazon EC2 instances that run your application.

Root volume (boot device)

Root volume type

The type of storage for the root volume attached to each instance.

(Container default)

Size

The number of gigabytes of the root volume attached to each instance.

GB

IOPS

Input/output operations per second for a provisioned IOPS (SSD) volume.

100

IOPS

Throughput

The desired throughput for the Amazon EBS root volume attached to your environment's EC2 instance

125

MiB/s

Amazon CloudWatch monitoring

The time interval between when metrics are reported from the EC2 instances

Monitoring interval

5 minute

Instance metadata service (IMDS)

Your environment's platform supports both IMDSv1 and IMDSv2. To enforce IMDSv2, deactivate IMDSv1. [Learn more](#)

IMDSv1

With the current setting, the environment enables only IMDSv2.

☒ Disable

EC2 security groups

Select security groups to control traffic.

EC2 security groups

Choose security groups

sg-06d80e515c32b9515 | default
default VPC security group



Clear all

Environment type: **Single instance**

Fleet composition: **On-demand instance**

Instance types: **t3.micro and t3.small** (choose the free tier available instance types)

Then only it can create an instance on demand. → Next.

Capacity [Info](#)

Configure the compute capacity of your environment and auto scaling settings to optimize the number of instances used.

Auto scaling group

Environment type

Select a single-instance or load-balanced environment. You can develop and test an application in a single-instance environment to save costs and then upgrade to a load-balanced environment when the application is ready for production. [Learn more](#)

Single instance

Fleet composition

Spot instances are launched at the lowest available price. [Learn more](#)

☒ On-Demand instance

☐ Spot instance

Architecture

The processor architecture determines the instance types that are made available. You can't change this selection after you create the environment. [Learn more](#)

☒ x86_64

This architecture uses x86 processors and is compatible with most third-party tools and libraries.

☐ arm64

This architecture uses AWS Graviton processors. You might have to recompile some third-party tools and libraries.

Instance types

Add instance types for your environment with your preferred launch order. The order preference only applies to On-Demand Instances and Spot Instances that use the capacity optimized prioritized allocation strategy. We recommend you include at least two instance types. [Learn more](#)

1. t3.micro



2. t3.small



Remove

Add instance type

AMI ID

Elastic Beanstalk selects a default Amazon Machine Image (AMI) for your environment based on the Region, platform version, and processor architecture that you choose. [Learn more](#)

ami-058121d1c7fafc3b9

Cancel

Skip to Review

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Next

Keep the log and monitoring as default → Next.

Amazon X-Ray

Amazon X-Ray is a service that collects data about the requests and responses that your application serves and receives. You can use the tools that X-Ray offers to view and filter the data that it provides to identify potential issues and optimization opportunities.

X-Ray daemon

service charges may apply.

☐ Enable

S3 log storage

Configure the instances in your environment to upload rotated logs to Amazon S3. [Learn more](#)

Rotate logs

standard S3 charges apply.

☐ Enable

Instance log streaming to CloudWatch logs

Configure the instances in your environment to stream logs to CloudWatch logs. You can set the retention to up to 10 years and configure Elastic Beanstalk to delete the logs when you terminate your environment. [Learn more](#)

Log streaming

standard CloudWatch charges apply.

☐ Enable

Environment properties

Use environment properties as either plain text environment variables or as references to secret ARN values that are stored in AWS Secrets Manager or AWS Systems Manager (SSM) Parameter Store. When using secrets, update your Elastic Beanstalk EC2 instance profile role with permissions to your secrets and parameters. [Learn more](#)

Source	Name	Value	
Plain text ▼	GRADLE_HOME	/usr/local/gradle	Remove
Plain text ▼	M2	/usr/local/apache-maven/bin	Remove
Plain text ▼	M2_HOME	/usr/local/apache-maven	Remove
Add environment property			

Cancel

[Previous](#)

Next

Review and click **create**.

Review

Step 1: Configure environment

Environment information

Environment tier

Web server environment

Environment name

Myapplication-env

Platform

arn:aws:elasticbeanstalk:ap-south-1::platform/Corretto 25 running on 64bit Amazon Linux 2023/4.11.1

Application name

Myapplication

Application code

Sample application

Step 2: Configure service access

Service access [Info](#)

Configure the service role and EC2 instance profile that Elastic Beanstalk uses to manage your environment. Choose an EC2 key pair to securely log into your instances.

Service role

```
arn:aws:iam::314383175192:role/aws-elasticbeanstalk-service-  
role
```

EC2 key pair

mumke

EC2 in:

aws-el;

Step 3: Set up networking, database, and tags

Step 5: Configure updates, monitoring, and logging

[Edit](#)

Updates, monitoring, and logging [Info](#)

Define when and how Elastic Beanstalk deploys changes to your environment. Manage your application's monitoring and logging settings, instances, and other environment resources.

Monitoring

System	Cloudwatch custom metrics - instance	Cloudwatch custom metrics - environment
enhanced	—	—
Log streaming	Retention	Lifecycle
Disabled	7	false

Updates

Managed updates	Deployment batch size	Deployment batch size type
Enabled	100	Percentage
Command timeout	Deployment policy	Health threshold
600	AllAtOnce	Ok
Ignore health check	Instance replacement	
false	false	

Platform software

Lifecycle	Log streaming	Logs retention
false	Disabled	7
Rotate logs	Update level	X-Ray enabled
Disabled	minor	Disabled

Environment properties

Source	Key	Value
Plain text	GRADLE_HOME	/usr/local/gradle
Plain text	M2	/usr/local/apache-maven/bin
Plain text	M2_HOME	/usr/local/apache-maven

[Cancel](#)[Previous](#)[Create](#)

Step no:7 Once created check its **health** is **OK** → click it → we can see a **DNS** for the EBS.

Environments (1) [Info](#)

Environment name	Health	Application name	Platform	Domain
Myapplication-env	Ok	Myapplication	Corretto 2...	Myapplication-env.eba-nanafq...

Myapplication-env [Info](#)

[Actions](#)[Upload and deploy](#)

Environment overview

Health
Pending

Environment ID
e-nq47pew5q5

Domain
[Myapplication-env.eba-nanafqtb.ap-south-1.elasticbeanstalk.com](#)

Application name
Myapplication

Platform

[Change version](#)

Platform
Corretto 25 running on 64bit Amazon Linux 2023/4.11.1

Running version
—

Platform state
Supported

[Events](#) [Deployments](#) [Health](#) [Logs](#) [Monitoring](#) [Alarms](#) [Managed updates](#) [Tags](#)

Events (10) [Info](#)

Time	Type	Details
April 15, 2026 17:45:27 (UTC+5:30)	INFO	Instance deployment completed successfully.
April 15, 2026 17:45:22 (UTC+5:30)	INFO	Added instance [i-0041aa3bfcf525bd3] to your environment.
April 15, 2026 17:45:20 (UTC+5:30)	INFO	Instance deployment used the commands in your 'Profile' to initiate startup of your application.
April 15, 2026 17:45:20 (UTC+5:30)	INFO	Instance deployment successfully built your application using commands in the 'Buildfile'.

Task: To connect a EC2 instance via SSM Session Manager

Step no:1 First create a public instance.

Name: Mywebserever.

Instances (2) [Info](#) Last updated less than a minute ago Connect Instance

Saved filter sets
 Choose filter set

Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	Myapplication-env	i-0041aa3bfcf525bd3	Terminated	t3.micro	–	View alarms +
☐	Mywebserver	i-035bf56df87fc7fe8	Running	t3.micro	3/3 checks passed	View alarms +

Step no:2 Once created Go to IAM → Roles → Create role.

IAM > Roles Delete Create role

Search IAM

Dashboard

Access Management

- User groups
- Users
- Roles**
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Roles (14) [Info](#) An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Search

☐	Role name	Trusted entities	Last activity
☐	aws-elasticbeanstalk-ec2-role	AWS Service: ec2	11 minutes ago
☐	aws-elasticbeanstalk-service-role	AWS Service: elasticbeanstalk	11 minutes ago
☐	AWSServiceRoleForAmazonElasticFileSystem	AWS Service: elasticfilesystem (Service-Linked)	14 days ago
☐	AWSServiceRoleForAutoScaling	AWS Service: autoscaling (Service-Linked)	12 minutes ago
☐	AWSServiceRoleForBackup	AWS Service: backup (Service-Linked)	7 hours ago
☐	AWSServiceRoleForElasticLoadBalancing	AWS Service: elasticloadbalancing (Service-Linked)	5 days ago
☐	AWSServiceRoleForNATGateway	AWS Service: ec2-nat-gateway (Service-Linked)	11 days ago

Entity type: AWS service

Use case: EC2

Next.

Select trusted entity [Info](#)

Trusted entity type

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.

Use case

☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.

☐ **EC2 Role for AWS Systems Manager**
Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.

☐ **EC2 Spot Fleet Role**
Allows EC2 Spot Fleet to request and terminate Spot Instances on your behalf.

☐ **EC2 - Spot Fleet Auto Scaling**
Allows Auto Scaling to access and update EC2 spot fleets on your behalf.

☐ **EC2 - Spot Fleet Tagging**
Allows EC2 to launch spot instances and attach tags to the launched instances on your behalf.

☐ **EC2 - Spot Instances**
Allows EC2 Spot Instances to launch and manage spot instances on your behalf.

☐ **EC2 - Spot Fleet**
Allows EC2 Spot Fleet to launch and manage spot fleet instances on your behalf.

☐ **EC2 - Scheduled Instances**
Allows EC2 Scheduled Instances to manage instances on your behalf.

Cancel Next

Permission → add permission → AmazonSSMManagedInstanceCore → next

Add permissions [Info](#)

Permissions policies (1/1145) [Info](#)

Choose one or more policies to attach to your new role.

Filter by Type		1 match
Q. AmazonSSMManagedInstanceCore	X	All types
Policy name i	Type	Description
☒ AmazonSSMManagedInstanceCore	AWS managed	The policy for Amazon EC2 Role to enable ...

► Set permissions boundary - optional

[Cancel](#) [Previous](#) [Next](#)

Provide a name for the policy: **Ec2-SSM-Role**

Role name

Enter a meaningful name to identify this role.

Ec2-SSM-Role

Maximum 64 characters. Use alphanumeric and '+=, @-/_' characters.

Description

Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-/_!#\$%^&*(){};:'"``

Step 1: Select trusted entities

Trust policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "ec2.amazonaws.com"
12        ]
13      }
14    ]
15  }
16 }
```

Step 2: Add permissions

Permissions policy summary

Policy name [i](#)

[AmazonSSMManagedInstanceCore](#)

▲ | Type

AWS managed

Step 3: Add tags

Add tags - optional [Info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create role](#)

Review → **Create role.**

Once created we can see the role.

 **Role Ec2-SSM-Role created.**

Roles (15) [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

 2 matches

☐	Role name	Trusted entities	Last activity
☐	aws-elasticbeanstalk-ec2-role	AWS Service: ec2	16 minutes ago
☐	Ec2-SSM-Role	AWS Service: ec2	-

Step no:3 Go our EC2 instance → **Mywebserver** → **Actions** → **Security** → **Modify IAM role**.

Instances (1/2) [Info](#)

Last updated less than a minute ago  [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[Saved filter sets](#) [Choose filter set](#)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☐	Myapplication-env	i-0041aa3bfcf525bd3	Terminated	t3.micro	-		
☒	Mywebserver	i-035bf56df87fc7fe8	Running	t3.micro	3/3 checks passed		

Change security groups

Modify IAM role

Get Windows password

Instance diagnostics

Instance settings

Networking

Security

Image and templates

Storage

Monitor and troubleshoot

i-035bf56df87fc7fe8 (Mywebserver)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary [Info](#)

Instance ID
 i-035bf56df87fc7fe8

IPv6 address
-

Hostname type
IP name: ip-172-31-38-144.ap-south-1.compute.internal

Public IPv4 address
 43.204.23.229 | [open address](#)

Instance state
 Running

Private IP DNS name (IPv4 only)
 ip-172-31-38-144.ap-south-1.compute.internal

Private IPv4 addresses
 172.31.38.144

Public DNS
 ec2-43-204-23-229.ap-south-1.compute.amazonaws.com | [open address](#)

Choose the role we have created for SSM connect: **Ec2-SSM-Role**.

Modify IAM role [Info](#)

Attach an IAM role to your instance.

Instance ID
 i-035bf56df87fc7fe8 (Mywebserver)

IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.



Once attached → **Reboot the instance.**

Then only the changes will be updated in the EC2.

✓ Successfully attached Ec2-SSM-Role to instance i-035bf56df87fc7fe8

Instances (1/2) [Info](#)

Saved filter sets
Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type
☐	Myapplication-env	i-0041aa3bfcf525bd3	Terminated	t3.micro
☒	Mywebserver	i-035bf56df87fc7fe8	Running	t3.micro

Step no:4 Now search **System manager** in the console → click it → Under System manger scroll down → managed nodes → click **Fleet manager**.

We can see our instance there. The status must be **online**. Then only it will work.

Also, once the IAM policy has been attached for the EC2 after rebooting wait for few minutes to get it reflected in the Fleet manager.

Systems Manager > Fleet Manager > Managed nodes

Fleet Manager [Info](#)

Settings ▼ Account management ▼

Managed Nodes (1)

Filter

Last fetched at: 6:24 PM

Node ID ▲

Node state ▼

Name ▼

Platform ... ▼

Operatin... ▼

Resource ... ▼

Source ID

Ping status ▼

Agent ver... ▼

Image ID ▼

EC2 insta...

i-035bf56df...

Running

Mywebserver

Linux

Amazon Linux

EC2 instance

-

Online

3.3.4108.0

ami-0e12ffc...

[Open EC2...](#)

Select the node instance → **Node actions** → **connect** → **Start terminal session.**

Fleet Manager [Info](#)

Settings ▾

Account management ▾

Managed Nodes (1)

Last fetched at: 6:27 PM

☒	Node ID ▲	Node state ▼	Name ▼	Platform ... ▼	Operatin... ▼	Resource ... ▼	Source ID	Ping status ▼	Agent ver... ▼	Image ID ▼	EC2 insta...
☒	i-035bf56df...	Running	Mywebserver	Linux	Amazon Linux	EC2 instance	-	Online	3.3.4108.0	ami-0e12ffc...	Open EC2...



Report

Node actions ▲

Node overview

Start terminal session ↗

Connect ▶

Tools ▶

Node settings ▶

Session ID: root-3efc9jt8gke5dq2l92pur4ueie [Shortcuts](#) Instance ID: i-035bf56df87fc7fe8 (Mywebserver)

Terminate

sh-5.2\$

Also we can connect from EC2 as well. Go to our **EC2 console** → Select our respective instance → click **connect** → **SSM Session Manager** → We can see the connect **enabled** → click **connect** → our instance will be connected.

Connect [Info](#)

Connect to an instance using the browser-based client.

Instance ID
i-035bf56df87fc7fe8 (Mywebserver)VPC ID
vpc-0538c231a226fe3c9Security groups
sg-06d80e515c32b9515 (default)IAM role
Ec2-SSM-Role

EC2 Instance Connect

SSM Session Manager

SSH client

EC2 serial console

SSM agent status

Introducing Systems Manager just-in-time node access
Move towards zero standing privileges by requiring operators to request access before remotely connecting to instances. [Learn more](#)

Try for free ↗ X

Ping status
OnlineLast ping time
Wed Apr 15 2026 18:24:33 GMT+0530 (India Standard Time)Session Manager connection status
ConnectedAgent version
3.3.4108.0

Session Manager usage:

- Connect to your instance without SSH keys, a bastion host, or opening any inbound ports.
- Sessions are secured using an AWS Key Management Service key.
- You can log session commands and details in an Amazon S3 bucket or CloudWatch Logs log group.
- Configure sessions on the Session Manager [Preferences](#) page.

Cancel

Connect

Note: We are not logged in as EC2 user. Instead as a **SSM user**. We will have all sudo /admin access that can be performed.

To switch to home page as a EC2-user use the cd command: **cd /home/ec2-user.**